

Deep Learning Models for Abdominal CT Organ Segmentation in Children: Development and Validation in Internal and Heterogeneous Public Datasets

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Pediatric Imaging • Original Research

Keywords

children, deep learning, liver, segmentation, spleen

Submitted: Jan 24, 2024

Revision requested: Feb 7, 2024

Revision received: Apr 1, 2024

Accepted: Apr 17, 2024

First published online: May 1, 2024

Version of record: Jul 17, 2024

[Supplemental content](#) is available for this article.

The authors declare that there are no disclosures relevant to the subject matter of this article.

BACKGROUND. Deep learning abdominal organ segmentation algorithms have shown excellent results in adults; validation in children is sparse.

OBJECTIVE. The purpose of this article is to develop and validate deep learning models for liver, spleen, and pancreas segmentation on pediatric CT examinations.

METHODS. This retrospective study developed and validated deep learning models for liver, spleen, and pancreas segmentation using 1731 CT examinations (1504 training, 221 testing), derived from three internal institutional pediatric (age ≤ 18 years) datasets ($n = 483$) and three public datasets comprising pediatric and adult examinations with various pathologies ($n = 1248$). Three deep learning model architectures (SegResNet, DynUNet, and SwinUNETR) from the Medical Open Network for Artificial Intelligence (MONAI) framework underwent training using native training (NT), relying solely on institutional datasets, and transfer learning (TL), incorporating pretraining on public datasets. For comparison, TotalSegmentator, a publicly available segmentation model, was applied to test data without further training. Segmentation performance was evaluated using mean Dice similarity coefficient (DSC), with manual segmentations as reference.

RESULTS. For internal pediatric data, the DSC for TotalSegmentator, NT models, and TL models for normal liver was 0.953, 0.964–0.965, and 0.965–0.966, respectively; for normal spleen, 0.914, 0.942–0.945, and 0.937–0.945; for normal pancreas, 0.733, 0.774–0.785, and 0.775–0.786; and for pancreas with pancreatitis, 0.703, 0.590–0.640, and 0.667–0.711. For public pediatric data, the DSC for TotalSegmentator, NT models, and TL models for liver was 0.952, 0.871–0.908, and 0.941–0.946, respectively; for spleen, 0.905, 0.771–0.827, and 0.897–0.926; and for pancreas, 0.700, 0.577–0.648, and 0.693–0.736. For public primarily adult data, the DSC for TotalSegmentator, NT models, and TL models for liver was 0.991, 0.633–0.750, and 0.926–0.952, respectively; for spleen, 0.983, 0.569–0.604, and 0.923–0.947; and for pancreas, 0.909, 0.148–0.241, and 0.699–0.775. The DynUNet TL model was selected as the best-performing NT or TL model considering DSC values across organs and test datasets and was made available as an open-source MONAI bundle (https://github.com/cchmc-dll/pediatric_abdominal_segmentation_bundle.git).

CONCLUSION. TL models trained on heterogeneous public datasets and fine-tuned using institutional pediatric data outperformed internal NT models and TotalSegmentator across internal and external pediatric test data. Segmentation performance was better in liver and spleen than in pancreas.

CLINICAL IMPACT. The selected model may be used for various volumetry applications in pediatric imaging.

Solid abdominal organ segmentation transforms raw image data into spatially structured information relevant to biomedical image analysis and potentially relevant to clinical decision-making and research. Organ volumes are potential biomarkers of disease that are objectively better quantitative representations of organ sizes than linear or multilinear measurements and may be more sensitive to change (i.e., disease progression or therapeutic effect) [1]. Furthermore, organ segmentation can be used to facilitate image analysis for biomarker application and validation, disease monitoring (e.g., Gaucher disease [2, 3], hepatic steatosis [4], or pancreatitis [5]), opportunistic screening [6], patient-specific dosimetry [7], nomogram generation [8], and input to computer-aided diagnostic models that rely on radiomic and/or deep learning features [9].

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doi.org/10.2214/AJR.24.30931

AJR 2024; 223:e2430931

ISSN-L 0361-803X/24/2231–e2430931

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AJR:223, July 2024

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Manual and semiautomated methods for medical image segmentation are generally time-consuming, impacted by interreader agreement, and difficult to accommodate into routine clinical workflows [10]. The time required for segmentation tasks also increases as the number of organs and tissues of interest increases (e.g., concurrent segmentation of liver, spleen, and pancreas).

Studies have shown that abdominal organ segmentation can be reliably performed using deep neural network and transformer models for various imaging modalities [10–16]. The TotalSegmentator model pipeline that segments 104 different organs on CT images was introduced in 2023 [17]. TotalSegmentator was developed using 1204 CT examinations showing a variety of pathologies that were acquired using varying imaging protocols and scanners from multiple manufacturers. However, the dataset contained only four pediatric examinations.

Even though existing models for performing whole-volume organ segmentation have shown excellent results in adults, their performance in children remains largely unexplored. Children differ from adults both in abdominal organ size and in the quantity of intraabdominal fat separating organs [18–20]. The effectiveness of automatic deep learning models in segmenting the pediatric pancreas is of particular concern given this organ's small size and variable morphology.

When developing and validating artificial intelligence (AI)-based segmentation models for clinical deployment, trade-offs exist between using institution-specific high-quality datasets versus using publicly available open-source models and datasets that may have organ segmentations of variable quality. Although models trained on large datasets from different institutions generally show consistent performance on new data, they must be validated for the intended institution's scanner settings. Additionally, a thorough evaluation of benchmark neural network architectures for organ segmentation and of advanced learning approaches (e.g., transfer learning [TL] and semisupervised learning with appropriate hyperparameter tuning) is essential to determine the optimal model for clinical use in terms of segmentation and computational performance.

This study's aim was to develop and validate deep learning models for simultaneous segmentation of the liver, spleen, and pancreas on pediatric abdominal CT examinations, using manual organ segmentations as the reference standard.

Methods

This HIPAA-compliant retrospective study was approved by the institutional review board. The requirement for participant written informed consent was waived.

Datasets

The study included abdominopelvic CT examinations from internal clinical datasets at a large tertiary pediatric academic medical center (Cincinnati Children's Hospital Medical Center) and from public segmentation datasets. Figure 1 is a flow diagram with details about sample selection criteria and the number of examinations used for training and testing in each dataset.

The internal CT examinations were obtained from three existing institutional datasets for which organs were previously segmented as part of prior projects related pediatric nomograms indicating age- and sex-specific organ volumes. The first internal

Highlights

Key Finding

- Deep learning models were developed for pediatric abdominal organ segmentation on CT by training using public datasets comprising adult and pediatric examinations with various pathologies, followed by fine-tuning in internal institutional pediatric data. The optimal model outperformed models trained using only internal pediatric data as well as a publicly available model.

Importance

- The best-performing organ segmentation model, based on internal and external validation, is now available as an open-source application for further research and potential clinical deployment.

dataset, LS-pediatric (of the liver and spleen), comprised 320 CT examinations performed from January 2018 to July 2021 in 320 children (age range, 2–18 years) in which the liver and spleen were previously manually segmented. The second internal dataset, P-pediatric (of the pancreas), comprised 140 CT examinations performed from January 2012 to July 2021 in 140 children (age range, 0–18 years) in which the pancreas was previously manually segmented. All examinations in both datasets were included in prior investigations establishing volumes for normal liver [19], normal spleen [19], or normal pancreas [13]. Both datasets were assembled using a similar iterative sampling and selection process with greedy search methods whereby patients who met predefined criteria were systematically identified until reaching a predetermined quota of 10 patients for each age-sex grouping (32 and 14 such groupings for LS-pediatric and P-pediatric datasets, respectively). The datasets avoided selection, based on review of the EHR and PACS, of patients with chronic disease; relevant laboratory abnormalities (P-pediatric dataset only); any of hepatosplenomegaly, jaundice, or lymphadenopathy (LS-pediatric dataset only); or prior pancreatic surgery (P-pediatric dataset only), as well as of examinations with visible abnormalities on CT of the organ of interest, incomplete coverage of the organ of interest, or motion artifacts. Figure 1 provides additional selection criteria.

The third internal dataset, P-pediatric-pancreatitis, was extracted from a larger internal registry of patients with acute pancreatitis or acute recurrent pancreatitis in which the pancreas was previously manually segmented. The P-pediatric-pancreatitis dataset included 23 CT examinations from the larger registry that were performed from January 2013 to July 2021 in 23 children (age range, 0–18 years) at least 30 days after the initial pancreatitis diagnosis. Six examinations from this dataset were included in a prior investigation of pancreatic volumes in children with pancreatitis [5].

In addition, three publicly available CT datasets for which the liver, spleen, and pancreas were previously manually segmented were used to pretrain models and to investigate models' generalizability. These datasets included the following: the [TotalSegmentator](#) version 1 dataset primarily in adults (age range, 16–96 years; $n = 1204$) [17]; [The Cancer Imaging Archive](#) (TCIA) pediatric dataset (age range, 0–16 years; $n = 359$) [21]; and the [Beyond the](#)

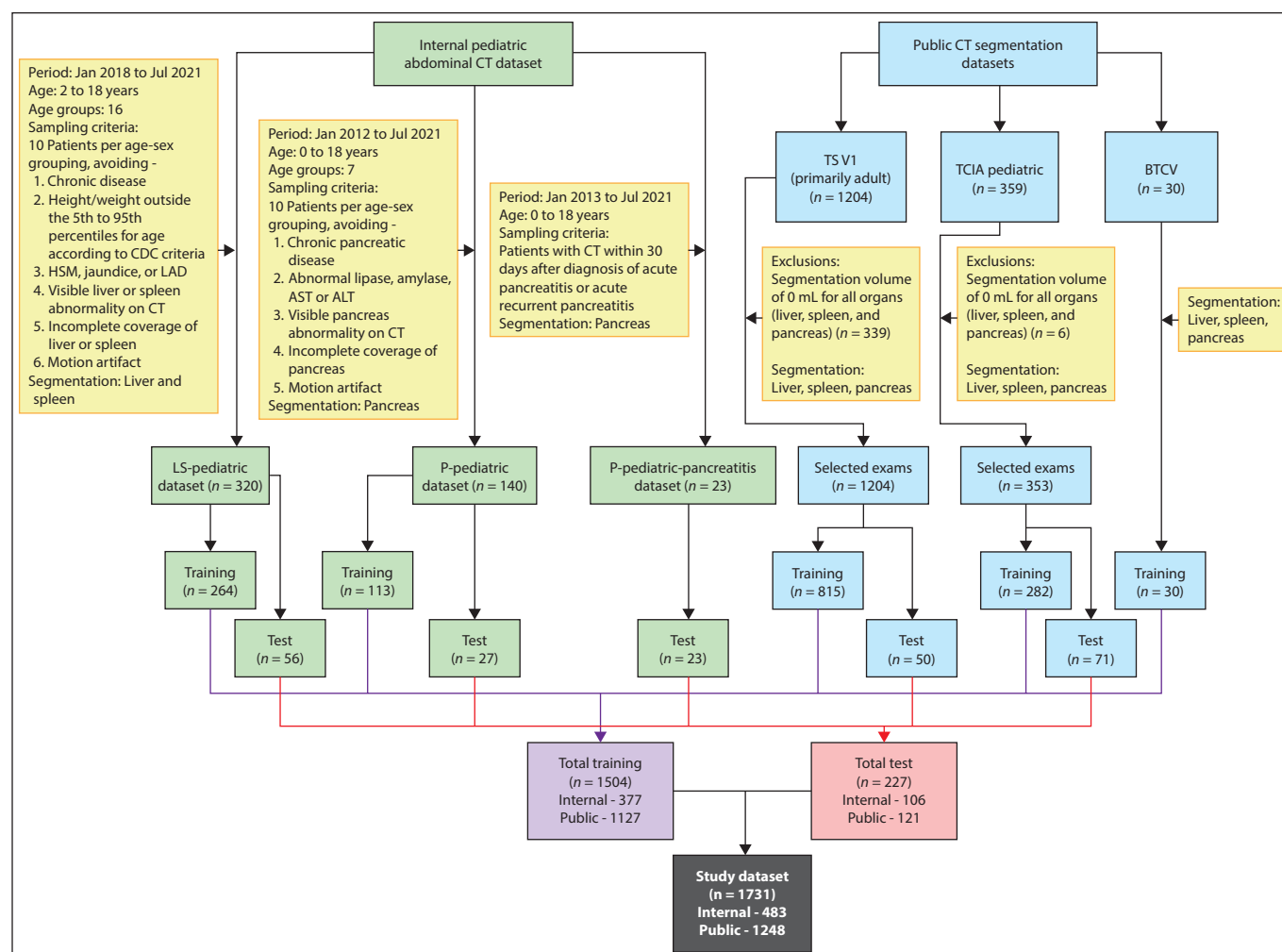


Fig. 1—Data flow diagram shows internal and public CT examinations used for model development and validation. HSM = hepatosplenomegaly, LAD = lymphadenopathy, AST = aspartate transaminase, ALT = alanine transaminase, V1 = version 1, TCIA = The Cancer Imaging Archive, BTCV = Beyond the Cranial Vault, LS = liver-spleen, P = pancreas.

Cranial Vault (BTCV) dataset (demographic information unavailable; $n = 30$) [22]. The three public datasets were derived from populations from heterogeneous geographic regions and with heterogeneous ranges of pathologies; patients in TCIA all had known cancer diagnoses. The acquisition periods were unknown for the public datasets. For the present investigation, examinations with a segmentation volume of 0 mL (e.g., due to the organ's absence in the FOV) for all three organs (liver, spleen, and pancreas) were excluded for the TotalSegmentator ($n = 339$) and TCIA ($n = 6$) datasets; no examinations were excluded on this basis for the BTCV dataset. Thus, the present investigation included 865, 353, and 30 examinations for TotalSegmentator, TCIA, and BTCV, respectively. All examinations from external datasets were treated as though derived from distinct patients.

LS-pediatric, P-pediatric, and TCIA datasets were split into training (i.e., pretraining and fine-tuning) and test datasets, using a random holdout of approximately 18% of cases in each dataset for testing. The P-pediatric-pancreatitis dataset was used only for testing. TotalSegmentator was split into training and test datasets based on the same split used in the model's original validation.

The BTCV dataset was used only for training. Thus, training and testing each used five distinct subsets of the study datasets, and no examination overlapped between training and testing data. Through this process, testing was performed on internal pediatric data for liver and spleen in the LS-pediatric dataset ($n = 56$) and for pancreas in the P-pediatric ($n = 27$) and P-pediatric-pancreatitis ($n = 23$) datasets; testing for liver, spleen, and pancreas was also performed on public TotalSegmentator primarily adult data ($n = 50$) and public TCIA pediatric data ($n = 71$). In total, across the six datasets, model development and validation used 1731 CT examinations (483 internal, 1248 public; training in 1504 [377 internal, 1127 public]; testing in 227 [106 internal, 121 public]).

CT Acquisition Techniques

CT examinations in the three internal datasets were performed using several scanners (AquilionOne or AquilionOne Genesis, both Canon Medical Systems; Revolution Apex, GE HealthCare) and were acquired in the portal venous phase after IV administration of iodinated contrast material. Images were reconstructed using standard iterative or deep learning techniques (AIDR3D or

AICE, both Canon Medical Systems; True Fidelity, GE HealthCare). The slice thickness was 3 mm for patients weighing less than 15 kg and 5 mm for patients weighing at least 15 kg; all examinations had no interslice gap. Spatial resolution varied, primarily related to patient age. The analysis used contrast-enhanced examinations because abdominal CT examinations at the institution were typically performed using IV contrast material and because contrast-enhanced examinations were considered to provide the best image quality for segmentation.

CT examinations from the three public datasets were performed using varying scanner types, acquisition parameters, and reconstruction techniques. These examinations varied in the use of IV contrast material. All examinations were reconstructed in the axial plane.

Manual Organ Segmentation in Internal Datasets

In the three internal datasets, software (3D Slicer, version 4.13.0) was used to initially create and subsequently review manual segmentations [23]. In the LS-pediatric dataset, the liver and spleen were previously manually segmented by a postdoctoral radiology research fellow (V.V.A.) with 1 year of experience in pediatric image analysis. In the P-pediatric dataset, the pancreas was previously segmented by a 4th-year medical student (L.Q.). In the P-pediatric-pancreatitis dataset, the pancreas was previously segmented by a clinical fellow in pediatric radiology (B.F.). Manual segmentation of a single organ typically required at least 15 minutes.

For the present investigation, two board-certified pediatric abdominal radiologists (J.R.D. and A.T.T., with 14 and 11 years of postfellowship experience, respectively) reviewed the prior manual segmentations, accessed through an institutional shared drive. The reviewers jointly assessed the segmentations and potentially edited segmentations after discussion. The reviewers' changes generally entailed minor adjustments near organ boundaries. The relative difference in organ volumes before and after this review was $0.0\% \pm 0.4\%$ (SD) for liver and $0.8\% \pm 2.6\%$ for spleen in the LS-pediatric dataset and $3.3\% \pm 4.1\%$ and $6.8\% \pm 9.5\%$ for pancreas in the P-pediatric and P-pediatric-pancreatitis datasets, respectively. These adjusted segmentations were used for all further analysis.

Deep Learning Models

First, the TotalSegmentator model pipeline [17], based on the nnU-Net architecture [16] and publicly available as a Python package, was applied to the five test datasets, without further training for the present investigation. The TotalSegmentator pipeline uses heuristic rules for preprocessing and hyperparameter selection based on the target dataset. It uses five model checkpoints from fivefold cross-validation to segment 104 target organs and applies largest connected component extraction to each organ segmentation, to enhance performance. The present investigation used the pipeline's high-resolution 1.5-mm 3D version.

Three top-performing model architectures for segmentation tasks from the Medical Open Network for Artificial Intelligence (MONAI) framework [24] underwent training (as described later in the Methods), followed by validation in the five test datasets. The MONAI framework offers open-source optimized deep learning libraries for medical image analysis, along with tool kits to facilitate models' clinical deployment. The first model architec-

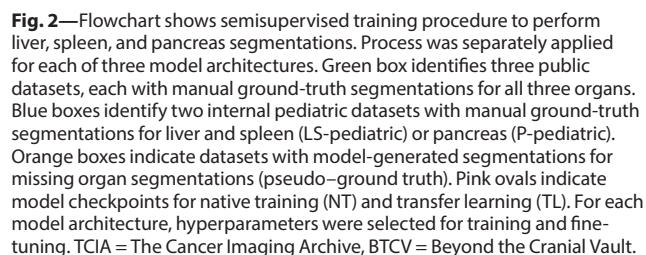
ture examined was [SegResNet](#), based on Resnet. SegResNet specializes in semantic segmentation and won the Multimodal Brain Tumor Segmentation (BRATS-18) challenge [25]. Compared with the symmetric U-Net [13] model, SegResNet has a more prominent encoder and smaller decoder. The present investigation did not use the original design's variational auto encoder. The second model architecture examined was [DynUNet](#), a fully convolutional neural network similar to the 3D U-Net [15] used by nnU-Net and that previously excelled in 19 medical image segmentation tasks [26]. DynUNet incorporates improvements over the original 3D U-Net, including instance normalization [27] and leakyReLU [28] activation functions. The third model architecture examined was [SwinUNETR](#), currently a benchmark in the BTCV [22] and BRATS-21 challenge [29]. SwinUNETR is a transformer-based segmentation model, using an image encoder from the Swin Transformer [14], which enhances the Vision Transformer [30] approach by efficiently calculating self-attention using shifted windows. The SwinUNETR's decoder mirrors the U-Net convolutional neural network design with skip connections. The three model architectures differed substantially in design, parameter size, and computational demands (see [Supplemental Methods](#) for more information).

Training Strategy

Figure 2 illustrates the study's approach for training the three model architectures. First, pretraining was performed using the three public datasets, all of which included segmentations for all three target organs. Further training was performed using two internal pediatric datasets—LS-pediatric (for liver and spleen segmentations) and P-pediatric (for pancreas segmentations). The pretrained models supplied pseudo-ground truths for missing organ segmentations in the internal datasets (e.g., missing pancreas segmentation in LS-pediatric dataset) to then develop the final models. This method enabled the merging of datasets from varied sources in a semisupervised training format without the need to manually perform missing segmentations. The process yielded two sets of models: those that were trained solely using the internal pediatric datasets (supplemented with pseudo-ground truths), which we called native-training (NT) models, and TL models that were fine-tuned using the internal pediatric datasets after pretraining using the public datasets. Each of the three model architectures underwent the training process separately, yielding separate NT and TL models for each architecture (i.e., three separate NT models and three separate TL models). For SwinUNETR, encoder initialization used self-supervised pretraining weights from the BTCV challenge; the other two model architectures started with random initialization.

Additional Training Details, Including Hyperparameter Tuning

For training the three model architectures, all examinations were resampled to a spatial resolution of $1.5 \times 1.5 \times 3.0$ mm³, using a patch size of $96 \times 96 \times 96$ pixels. Images underwent thresholding from -250 to 400 HU and were scaled from 0 to 1. During training, eight patches were randomly sampled, stratified across four classes (liver, spleen, pancreas, and background) from each examination for every batch iteration. A total of 15% of the training cases were set aside as a tuning dataset to monitor training and select the best model. For all architectures, training was per-



Training of NT and TL models incorporated hyperparameter tuning to determine the best model parameters. This tuning entailed a grid search on 50 randomly selected examinations for 50 epochs. The process tested learning rates ranging from 1×10^{-5} to 5×10^{-3} , as well as different learning rate schedulers, including linear, cosine, linear warm-up cosine, and step. The hyperparameters yielding the best Dice similarity coefficient

For training, the combined DSC and cross-entropy loss function [31] was optimized using the Adam algorithm with a weight decay of 0.001. Automatic mixed precision was used for efficiency and regularization. On-the-fly data augmentations, such as random rotations and intensity shifts of up to 5%, were applied to half of the training patches. Models were trained for 200 epochs regularly and for 150 epochs during fine-tuning. The batch size was adjusted for each architecture based on GPU memory. During inference, a sliding window overlap factor of 0.75 and batch size of four were used.

Models' segmentation performance in each test dataset was assessed using the mean DSC and mean volumetric similarity (VS) [32]. DSC measured overlap between predicted and ground-truth segmentations, ranging from 0 (no overlap) to 1 (exact match). VS measured alignment between predicted and ground-truth segmentations, ranging from -2 (positive predicted volume in presence of ground-truth volume of zero) to 2 (predicted volume of zero in presence of positive predicted volume), with a score of 0 indicating an exact match. A single best NT or TL model was selected considering DSC values across organs and test datasets. DSC and VS were compared between the selected model and TotalSegmentator in the test datasets. These assessments were performed using the [SciPy](#) package [33].

Processing Times

Results

Datasets

Table 1 summarizes demographic characteristics and volumes of manually segmented organs for each dataset (LS-pediatric [mean age, 10 years; 160 boys, 160 girls]; P-pediatric [mean age, 8 years; 70 boys, 70 girls]; P-pediatric-pancreatitis [mean age, 11 years; 13 boys, 10 girls]; TotalSegmentator [mean age, 64 years; 512 men, 353 women]; TCIA [mean age, 7 years; 159 boys, 194 girls]; BTCV [information unavailable]). The recorded volume was

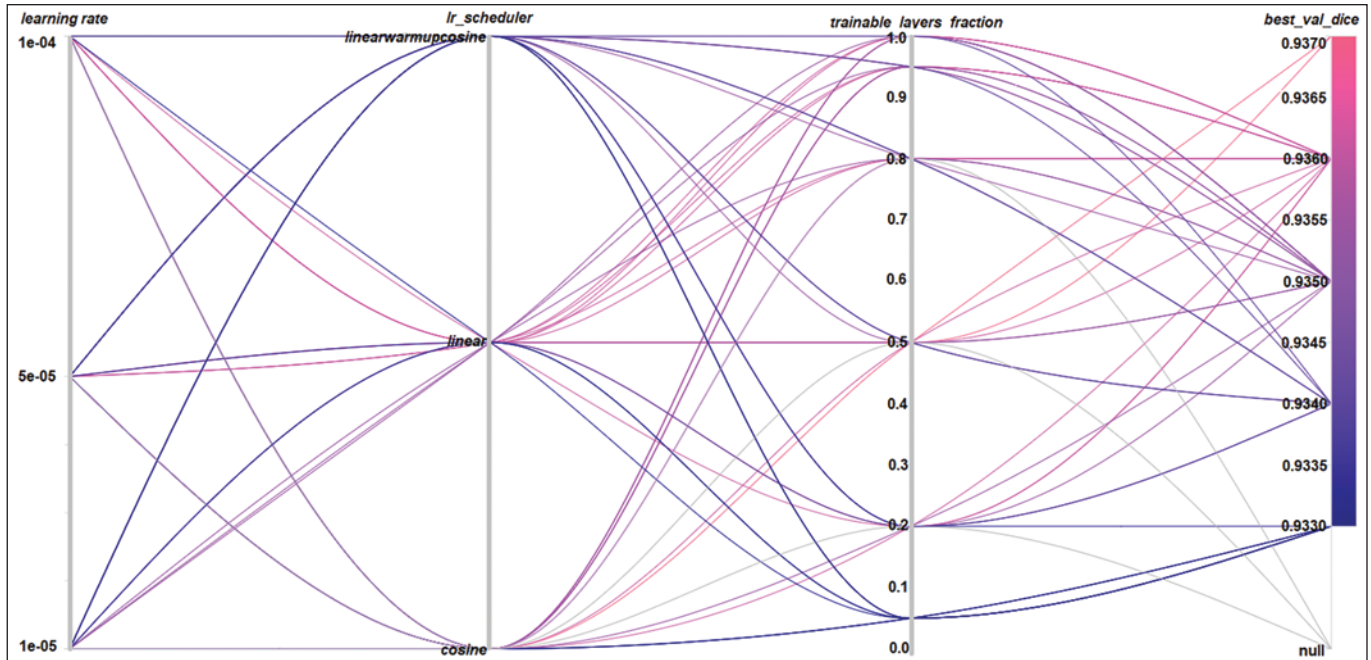


Fig. 3—Graphic shows parallel coordinates plot for hyperparameter search to fine-tune DynUNet (Medical Open Network for Artificial Intelligence) transfer learning model on liver and spleen pediatric dataset. Search was performed to optimize initial learning rate (lr), lr_scheduler, and fraction of trainable layers in pretrained model, based on validation Dice similarity coefficient (DSC). Combination of parameters that produced best validation DSC (best_val_dice) after 50 epochs when trained on random subset of 50 examinations was used for fine-tuning final model. Model was unstable for certain parameter combinations, resulting in null values.

0 mL for the pancreas in 98 examinations in TotalSegmentator and 11 examinations in TCIA, and for the spleen in one examination in TotalSegmentator.

Performance on Internal Pediatric Test Data

Table 2 shows segmentation performance for normal liver and normal spleen in the LS-pediatric dataset. For liver, mean DSC was 0.953 for TotalSegmentator, ranged from 0.964 to 0.965 for NT models, and ranged from 0.965 to 0.966 for TL models. For spleen, mean DSC was 0.914 for TotalSegmentator, ranged from 0.942 to 0.945 for NT models, and ranged from 0.937 to 0.945 for TL models. For liver, mean VS was 0.016 for TotalSegmentator, ranged from -0.009 to 0.012 for NT models, and ranged from -0.006 to 0.001 for TL models. For spleen, mean VS was 0.070 for TotalSegmentator, ranged from -0.013 to -0.002 for NT models, and ranged from -0.018 to -0.008 for TL models.

Table 3 shows segmentation performance for normal pancreas in the P-pediatric dataset and for pancreas with pancreatitis in the P-pediatric-pancreatitis dataset. For normal pancreas, mean DSC was 0.733 for TotalSegmentator, ranged from 0.774 to 0.785 for NT models, and ranged from 0.775 to 0.786 for TL models. For pancreas with pancreatitis, mean DSC was 0.703 for TotalSegmentator, ranged from 0.590 to 0.640 for NT models, and ranged from 0.667 to 0.711 for TL models. For normal pancreas, mean VS was 0.073 for TotalSegmentator, ranged from -0.003 to 0.026 for NT models, and ranged from -0.020 to 0.042 for TL models. For pancreas with pancreatitis, mean VS was 0.022 for TotalSegmentator, ranged from 0.326 to 0.513 for NT models, and ranged from 0.168 to 0.266 for TL models.

Performance on TCIA Pediatric Test Data

Table 4 shows segmentation performance for liver, spleen, and pancreas in TCIA pediatric test data. Mean DSC for liver was 0.952 for TotalSegmentator, ranged from 0.871 to 0.908 for NT models, and ranged from 0.941 to 0.946 for TL models; for spleen was 0.905 for TotalSegmentator, ranged from 0.771 to 0.827 for NT models, and ranged from 0.897 to 0.926 for TL models; and for pancreas was 0.700 for TotalSegmentator, ranged from 0.577 to 0.648 for NT models, and ranged from 0.693 to 0.736 for TL models. Mean VS for liver was 0.039 for TotalSegmentator, ranged from 0.004 to 0.205 for NT models, and ranged from 0.055 to 0.060 for TL models; for spleen was 0.085 for TotalSegmentator, ranged from 0.111 to 0.327 for NT models, and ranged from -0.002 to 0.066 for TL models; and for pancreas was 0.140 for TotalSegmentator, ranged from 0.291 to 0.497 for NT models, and ranged from 0.208 to 0.293 for TL models.

Performance on TotalSegmentator Primarily Adult Test Data

Table 5 shows segmentation performance for liver, spleen, and pancreas in TotalSegmentator primarily adult data. Mean DSC for liver was 0.991 for TotalSegmentator, ranged from 0.633 to 0.750 for NT models, and ranged from 0.926 to 0.952 for TL models; for spleen was 0.983 for TotalSegmentator, ranged from 0.569 to 0.604 for NT models, and ranged from 0.923 to 0.947 for TL models; and for pancreas was 0.909 for TotalSegmentator, ranged from 0.148 to 0.241 for NT models, and ranged from 0.699 to 0.775 for TL models. Mean VS for liver was -0.002 for TotalSegmentator, ranged from 0.114 to 0.666 for NT models, and ranged from 0.055 to 0.106 for TL models;

TABLE 1: Details of the Patient Datasets Used for Model Development and Testing

Characteristic and Data Subset	Internal			Public ^a		
	LS-Pediatric	P-Pediatric	P-Pediatric-Pancreatitis	TotalSegmentator	TCIA Pediatric	BTCV
Examinations (no.)						
Total	320	140	23	865	353	30
Training	264	113	—	815	282	30
Test	56	27	23	50	71	—
Sex (no.)						
Male						
Total	160	70	13	512	159	NA
Training	132	54	—	353	123	
Test	28	16	13	29	36	
Female						
Total	160	70	10	353	194	NA
Training	132	59	—	332	159	
Test	28	11	10	21	35	
Age (y) ^b						
Total	10 (2–18)	8 (0–18)	11 (3–18)	64 (16–96)	7 (0–16)	NA
Training	10 (2–18)	8 (0–18)	—	64 (16–96)	7 (0–16)	
Test	10 (2–18)	9 (1–17)	11 (3–18)	63 (19–94)	7 (0–16)	
Liver volume (mL) ^b						
Total	910 (115–2002)	—	—	1385 (6–4164)	1002 (176–5906)	1739 (1033–2564)
Training	906 (115–1957)	—	—	1380 (6–4164)	973 (192–3904)	1739 (1033–2564)
Test	926 (417–2002)	—	—	1460 (711–2882)	1117 (176–5906)	—
Spleen volume (mL) ^b						
Total	152 (21–480)	—	—	210 (0–2544)	176 (4–1480)	339 (63–3087)
Training	152 (21–480)	—	—	207 (0–2544)	170 (4–1480)	339 (63–3087)
Test	152 (28–419)	—	—	247 (43–787)	200 (3–815)	—
Pancreas volume (mL) ^b						
Total	—	34 (3–105)	6 (13–90)	52 (0–146)	35 (0–186)	83 (41–138)
Training	—	33 (3–105)	—	52 (0–146)	34 (0–138)	83 (41–138)
Test	—	38 (13–68)	6 (13–90)	56 (0–106)	38 (0–186)	—

Note—Cells contain dash (—) when dataset did not include segmentation of given organ or did not include training or test data. LS = liver-spleen, P = pancreas, TCIA = The Cancer Imaging Archive, BTCV = Beyond the Cranial Vault, NA = not available.

^aVolume of 0 mL may be due to missing manual segmentation or absence of organ in examination's FOV.

^bReported as mean with range in parentheses.

for spleen was 0.006 for TotalSegmentator, ranged from 0.032 to 0.589 for NT models, and ranged from 0.015 to 0.101 for TL models; and for pancreas was –0.002 for TotalSegmentator, ranged from 0.316 to 0.828 for NT models, and ranged from 0.128 to 0.327 for TL models.

Processing Times

Table S1 shows the mean processing time per examination for different model architectures to segment the liver, spleen, and pancreas in the TotalSegmentator test data. The mean processing time was 8.6 seconds for SwinUNETR, 2.8 seconds for Seg-

ResNet, and 2.9 seconds for DynUNet. In comparison, for TotalSegmentator, the mean time per examination to segment 104 organs was 101.1 seconds.

Selection of Best Model

The DynUNet TL model was selected as the best model among NT and TL models across organs and test datasets. Among NT and TL models, this model had the highest DSC (or equaled the highest DSC) for normal liver (0.966), normal spleen (0.945), normal pancreas (0.786), and pancreas with pancreatitis (0.711) in the internal pediatric test data, as well as for spleen (0.926) and pan-

TABLE 2: Model Performance for Segmentation of Normal Liver and Normal Spleen in 56 Examinations in LS-Pediatric Test Data

Model	Liver		Spleen	
	DSC	VS	DSC	VS
TotalSegmentator	0.953 ± 0.006	0.016 ± 0.019	0.914 ± 0.039	0.070 ± 0.079
Native training				
SegResNet	0.965 ± 0.005	−0.009 ± 0.016	0.943 ± 0.015	−0.002 ± 0.038
DynUNet	0.965 ± 0.005	−0.001 ± 0.016	0.945 ± 0.015	−0.013 ± 0.034
SwinUNETR	0.964 ± 0.005	0.012 ± 0.018	0.942 ± 0.020	−0.006 ± 0.046
Transfer learning				
SegResNet	0.965 ± 0.005	−0.003 ± 0.017	0.943 ± 0.014	−0.018 ± 0.033
DynUNet	0.966 ± 0.005	−0.006 ± 0.017	0.945 ± 0.014	−0.008 ± 0.029
SwinUNETR	0.966 ± 0.004	0.001 ± 0.015	0.937 ± 0.053	−0.015 ± 0.112

Note—Data are mean ± SD. The model or models with the best score for each metric is highlighted in boldface. SegResNet, DynUNet, and SwinUNETR are model architectures from the Medical Open Network for Artificial Intelligence (MONAI) framework. LS = liver-spleen, DSC = Dice similarity coefficient, VS = volumetric similarity.

TABLE 3: Model Performance for Segmentation of Normal Pancreas and of Pancreas With Pancreatitis in P-Pediatric and P-Pediatric-Pancreatitis Test Data

Model	P-Pediatric Dataset (n = 27 Examinations)		P-Pediatric-Pancreatitis Dataset (n = 23 Examinations)	
	DSC	VS	DSC	VS
TotalSegmentator	0.733 ± 0.086	0.073 ± 0.153	0.703 ± 0.123	0.022 ± 0.350
Native training				
SegResNet	0.776 ± 0.068	−0.003 ± 0.187	0.640 ± 0.183	0.326 ± 0.500
DynUNet	0.785 ± 0.054	0.026 ± 0.165	0.590 ± 0.193	0.513 ± 0.521
SwinUNETR	0.774 ± 0.069	−0.008 ± 0.208	0.596 ± 0.199	0.362 ± 0.517
Transfer learning				
SegResNet	0.775 ± 0.052	0.042 ± 0.183	0.667 ± 0.160	0.251 ± 0.407
DynUNet	0.786 ± 0.055	−0.020 ± 0.171	0.711 ± 0.139	0.168 ± 0.317
SwinUNETR	0.783 ± 0.056	0.031 ± 0.187	0.675 ± 0.150	0.266 ± 0.326

Note—Data are mean ± SD. The model with the best score for each metric is highlighted in boldface. SegResNet, DynUNet, and SwinUNETR are model architectures from the Medical Open Network for Artificial Intelligence (MONAI) framework. P = pancreas, DSC = Dice similarity coefficient, VS = volumetric similarity.

creas (0.736) in the TCIA pediatric test data. This model was made available as an open-source MONAI [bundle](#) [34].

Table S2 summarizes comparisons of segmentation performance between the DynUNet TL and TotalSegmentator models. Figure 4 shows volume renderings of organs based on manual ground-truth segmentations as well as on segmentations by the DynUNet TL and TotalSegmentator models.

Discussion

To our knowledge, this is the first investigation to systematically assess a variety of deep learning architectures for abdominal organ segmentation in a large sample of internal pediatric and public CT datasets. TL approaches, whereby models were pre-trained on publicly available adult and pediatric data and then fine-tuned to the target institution's internal pediatric data, provided the best segmentation performance, outperforming the

TotalSegmentator model (without further training) as well as NT models (trained using only internal data). The TL models exhibited robust performance on internal (P-pediatric-pancreatitis) test data as well as on public pediatric (TCIA) and primarily adult (TotalSegmentator) test data. The NT models trained only on internal CT data showed similar performance to the TL models on normal organs from internal pediatric test data (LS-pediatric and P-pediatric datasets), but performed substantially worse on the internal pancreatitis test data (P-pediatric-pancreatitis dataset) and on the public test data. The findings support the utility of TL through pretraining on large public datasets for increasing model generalizability.

A range of factors likely contributed to the superior performance for TL models. The public datasets used for TL comprised an overall substantially larger number of cases than were available internally for NT. Additionally, the examinations from the

TABLE 4: Model Performance for Segmentation of Liver, Spleen, and Pancreas in 71 Examinations in TCIA Pediatric Test Set

Model	Liver		Spleen		Pancreas	
	DSC	VS	DSC	VS	DSC	VS
TotalSegmentator	0.952 ± 0.029	0.039 ± 0.051	0.905 ± 0.141	0.085 ± 0.263	0.700 ± 0.227	0.140 ± 0.255
Native training						
SegResNet	0.908 ± 0.175	0.004 ± 0.383	0.826 ± 0.245	0.111 ± 0.576	0.648 ± 0.264	0.291 ± 0.681
DynUNet	0.871 ± 0.205	0.205 ± 0.424	0.827 ± 0.221	0.188 ± 0.492	0.602 ± 0.260	0.482 ± 0.632
SwinUNETR	0.876 ± 0.198	0.146 ± 0.428	0.771 ± 0.315	0.327 ± 0.682	0.577 ± 0.272	0.497 ± 0.689
Transfer learning						
SegResNet	0.945 ± 0.048	0.055 ± 0.099	0.922 ± 0.121	-0.002 ± 0.248	0.707 ± 0.196	0.267 ± 0.422
DynUNet	0.941 ± 0.091	0.059 ± 0.188	0.926 ± 0.119	0.009 ± 0.248	0.736 ± 0.193	0.208 ± 0.406
SwinUNETR	0.946 ± 0.054	0.060 ± 0.109	0.897 ± 0.172	0.066 ± 0.373	0.693 ± 0.190	0.293 ± 0.397

Note—Data are mean ± SD. The model with the best score for each metric is highlighted in boldface. SegResNet, DynUNet, and SwinUNETR are model architectures from the Medical Open Network for Artificial Intelligence (MONAI) framework. TCIA = The Cancer Imaging Archive, DSC = Dice similarity coefficient, VS = volumetric similarity.

TABLE 5: Model Performance for Segmentation of Liver, Spleen, and Pancreas in 50 Examinations in the TotalSegmentator Primarily Adult Test Data

Model	Liver		Spleen		Pancreas	
	DSC	VS	DSC	VS	DSC	VS
TotalSegmentator	0.991 ± 0.006	-0.002 ± 0.012	0.983 ± 0.024	0.006 ± 0.044	0.909 ± 0.139	-0.002 ± 0.298
Native training						
SegResNet	0.750 ± 0.349	0.114 ± 0.847	0.595 ± 0.408	0.032 ± 1.136	0.241 ± 0.315	0.526 ± 1.547
DynUNet	0.668 ± 0.362	0.520 ± 0.806	0.569 ± 0.386	0.515 ± 0.810	0.148 ± 0.255	0.316 ± 1.741
SwinUNet	0.633 ± 0.378	0.666 ± 0.791	0.604 ± 0.388	0.589 ± 0.883	0.179 ± 0.259	0.828 ± 1.440
Transfer learning						
SegResNet	0.926 ± 0.164	0.106 ± 0.337	0.947 ± 0.040	0.015 ± 0.076	0.727 ± 0.235	0.281 ± 0.558
DynUNet	0.952 ± 0.099	0.055 ± 0.204	0.947 ± 0.056	0.043 ± 0.109	0.775 ± 0.239	0.128 ± 0.608
SwinUNet	0.942 ± 0.138	0.079 ± 0.283	0.923 ± 0.147	0.101 ± 0.293	0.699 ± 0.253	0.327 ± 0.629

Note—Data are mean ± SD. The model with the best score for each metric is highlighted in boldface. SegResNet, DynUNet, and SwinUNETR are model architectures from the Medical Open Network for Artificial Intelligence (MONAI) framework. DSC = Dice similarity coefficient, VS = volumetric similarity.

public datasets exhibited a range of pathologies and were obtained using heterogeneous acquisition techniques. In particular, the public datasets comprised both noncontrast and contrast-enhanced examinations, whereas the internal datasets used for NT comprised only contrast-enhanced examinations.

The DynUNet TL model was selected as the overall best-performing model in terms of performance across organs and datasets. This model was made available as an open-source MONAI bundle to facilitate future research and potential clinical deployment. Nonetheless, institutions applying the model should establish AI governance and monitoring infrastructure before deployment.

Liver and spleen segmentation performance on internal pediatric test data was better for the NT and TL models than for the publicly available TotalSegmentator model pipeline. This finding suggests that models trained on public datasets may be suboptimal for pediatric patients without fine-tuning on pediatric data. Despite this difference, the TotalSegmentator model achieved a DSC greater than 0.900 for both normal liver and normal spleen

on internal pediatric test data. On the other hand, the TotalSegmentator pipeline and three TL models had similar segmentation performance in the TCIA pediatric test data. The TotalSegmentator pipeline uses various components (cross-validation and largest connected component selection) to enhance segmentation robustness. Incorporating these components into the NT and TL model pipelines could further improve their performance. To our knowledge, the performance of automatic segmentation models for liver and spleen has previously been reported only in adults [34–37], yielding DSCs ranging from 0.950 to 0.983 and aligning with the present results in children. For comparison, an additional study reported a DSC of 0.964 for spleen segmentations between human observers [38].

Pancreatic segmentation on CT images remains challenging. The TotalSegmentator model had the best mean DSC (0.909) for the pancreas in the TotalSegmentator primarily adult test data. In contrast, the DynUNet TL model achieved the best mean DSCs in internal pediatric test data for both normal pancreas (0.786)

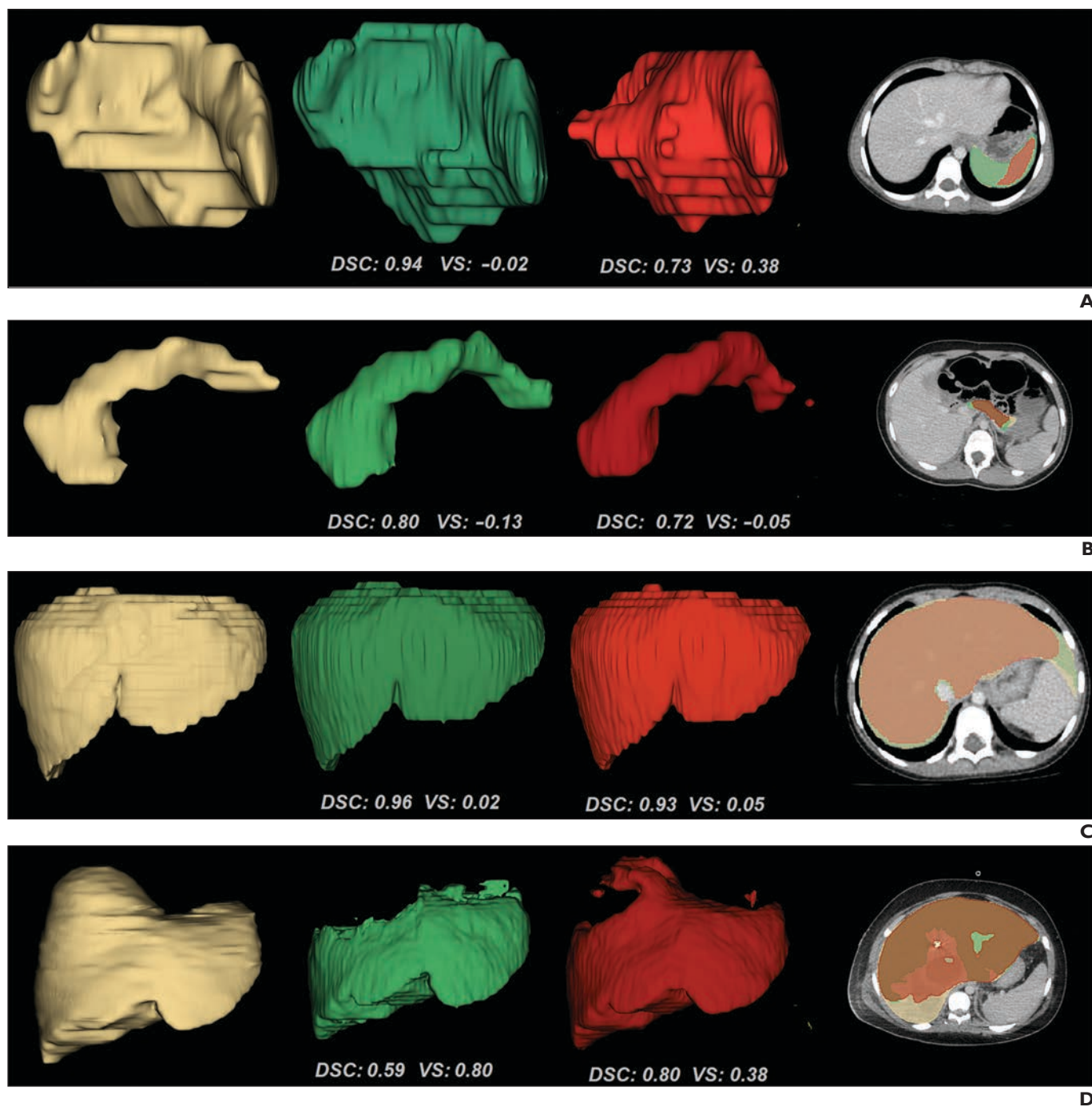


Fig. 4—Comparison of ground-truth manual organ segmentations (yellow), segmentations predicted by DynUNet (Medical Open Network for Artificial Intelligence) transfer learning (TL) (green), and TotalSegmentator (red) models in four patients. Internal CT examinations were obtained from existing institutional datasets for which organs were previously segmented: LS-pediatric (of liver and spleen) and P-pediatric (of pancreas). In each patient, final image is representative axial CT slice, showing segmentations. DSC = Dice similarity coefficient, VS = volumetric similarity.

A, Segmentation of normal spleen in 5-year-old girl from LS-pediatric test set shows performance was better for DynUNet TL than for TotalSegmentator model (DSC: 0.94 vs 0.73; VS: -0.02 vs 0.38).

B, Segmentation of normal pancreas in 11-year-old girl from P-pediatric test set shows performance was similar between DynUNet TL and TotalSegmentator model (DSC: 0.80 vs 0.72; VS: -0.13 vs -0.05).

C, Segmentation of normal liver in 6-year-old girl from LS-pediatric test set shows performance was similar between DynUNet TL and TotalSegmentator model (DSC: 0.96 vs 0.93; VS: 0.02 vs 0.05).

D, Segmentation of abnormal liver in 15-year-old boy with large liver tumor from The Cancer Imaging Archive pediatric test set shows performance was worse for DynUNet TL than for TotalSegmentator model (DSC: 0.59 vs 0.80; VS: 0.80 vs 0.38).

and pancreatitis (0.711), as well as in the TCIA pediatric test data (0.736). Earlier studies in adults have reported DSCs in the pancreas of 0.830 [39], 0.877 [40], and 0.910 [41]. An additional study also reported a DSC in the pancreas between human readers of 0.850 [41]. Pancreas segmentation may be particularly difficult in children because of the relatively small amount of peripancreatic retroperitoneal fat.

Manual segmentation of abdominal organs on CT images is a tedious process that is time-consuming for each individual organ segmented. The mean processing time per examination was 2.9 seconds for this study's selected model (DynUNet TL) vs 101.1 seconds (to segment 104 organs) for the TotalSegmentator model in the TotalSegmentator test dataset. This improved efficiency of the proposed model may facilitate clinical workflows in settings with limited computational resources. The ability to perform segmentation tasks in seconds can allow volumetry information to be available to radiologists at the time of interpretation. This quantitative information, when combined with reference nomograms of organ volumes, may aid early diagnosis of abdominal disease and facilitate downstream applications that require organ contours (e.g., radiomics-based predictions or patient-specific organ dosimetry).

This study had limitations. First, the internal pediatric test data did not include examinations with abnormalities of the liver or spleen. Nonetheless, we gauged the models' generalizability in an internal pancreatitis dataset and in two publicly available datasets (one pediatric and one primarily adult) of patients with a range of pathologies. Second, although the public datasets included patients with various pathologies, the exact distributions of pathologic conditions in the public datasets were unknown. Finally, we compared models only in terms of segmentation performance. Further investigations are warranted to assess the models' potential benefits or cost implications in real-world clinical settings.

In conclusion, pediatric-specific abdominal organ CT segmentation models were crafted by integrating multiple heterogeneous public datasets with local institutional examinations. TL (i.e., combining training from public data with model fine-tuning for the target population) yielded optimal performance. Systematic hyperparameter tuning yielded model architectures with distinct parameters but similar segmentation performance. Segmentation performance was overall better for the liver and spleen; pancreas segmentation remained challenging, even for the best-performing model architecture. The best-performing model was made publicly available as an open-source MONAI bundle for further research and potential clinical deployment for various volumetry applications in pediatric imaging.

Provenance and review: Not solicited; externally peer reviewed.

Peer reviewers: Maria Pilar Aparisi Gomez, Te Toka Tumai Auckland; Edward J. Richer, Emory + CHOA Pediatric Institute; Domen Plut, University Medical Centre Ljubljana; Houchun Harry Hu, University of Colorado; additional individual(s) who chose not to disclose their identity.

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(Editorial Comment starts on next page)

Editorial Comment: Using Appropriate Training Data in Deep Learning Tissue and Organ Segmentations on CT

Text quantification of the volume of tissues and organs via segmentation of radiologic images is a fundamental process in clinical research [1, 2]. CT, with image pixels represented by well-defined Hounsfield units, provides a natural and efficient image processing platform for gray-scale intensity-based segmentation. In recent years, deep learning (DL) and artificial intelligence have rapidly reduced the time needed to segment large volumes of CT data to just a few seconds.

In this study, the authors train and validate their CT-based DL model for segmenting the liver, spleen, and pancreas in children [3]. In the spirit of reproducible research, a must in DL-based medical imaging research, the authors make available to the community their most optimal algorithm via the Medical Open Network for Artificial Intelligence (MONAI) platform.

An Achilles heel of DL-based image segmentation is selection bias in training and validation datasets [4], which can lead to inadvertent results. End users of DL-based segmentation pipelines are encouraged to be cognizant of the data used in training and validating such algorithms and to be critical of the results. Equally, developers of such pipelines and contributors of data to public databases are encouraged to be transparent with the origins and details of the involved data (e.g., data acquisition modality settings, known preprocessing steps, patient demographics).

The study's authors make a commendable effort to improve their work's generalizability by supplementing pediatric CT datasets from their own practice with three publicly available databases that encompass a variety of pathologies and heterogeneous

demographics. Chosen with care and assessed with a study design that includes k-fold cross-validation, the authors show that a pipeline incorporating training with data from the public databases can meaningfully augment segmentation accuracy.

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First published online: May 1, 2024 • Version of record: Jul 17, 2024

The author declares that there are no disclosures relevant to the subject matter of this article.

doi.org/10.2214/AJR.24.31345 • AJR 2024; 223:e2431345

ISSN-L 0361-803X/24/2231–e2431345 • © American Roentgen Ray Society

Provenance and review: Solicited; not externally peer reviewed.

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(Editorial Comment starts on next page)

Editorial Comment: Usefulness of a Deep Learning Model for Pediatric Abdominal Organ Segmentation

Research and advances in deep learning and artificial intelligence (AI) for applications relevant to radiology have become an important focus in recent years. Much of that spotlight, however, has shone on applications for adult imaging, with relatively little attention given to pediatric radiology [1]. To wit, one of the large publicly available organ segmentation datasets used in this study [2] (TotalSegmentator) contained only four pediatric examinations in a total dataset of 1204 examinations.

The authors of this study have made a commendable effort to narrow that attention gap through the creation of a deep learning model that accurately segments the pediatric liver, spleen, and pancreas, and they have made that model available as open-source software. This model can facilitate both research and patient care by accelerating what would normally be a tedious and time-consuming process to calculate organ volumes. Easily obtainable organ volume measurements could yield more precise research and more finely tuned clinical patient monitoring and treatment of a wide variety of conditions [3].

Additional research and validation of the model with larger datasets, datasets from additional institutions, and datasets obtained in the setting of a variety of organ pathologies is certainly warranted. The availability of the model through the open-source Medical Open Network for Artificial Intelligence (MONAI) framework will facilitate that validation and potentially encourage further innovation in pediatric AI applications [4]. As we move toward a future of radiology ever more intertwined with

AI, the availability of reliable AI tools for all patients, both adult and pediatric, is crucial.

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First published online: May 15, 2024 ■ Version of record: Jul 17, 2024

The author declares that there are no disclosures relevant to the subject matter of this article.

doi.org/10.2214/AJR.24.31408 ■ AJR 2024; 223:e2431408

ISSN-L 0361-803X/24/2231–e2431408 ■ © American Roentgen Ray Society

Provenance and review: Solicited; not externally peer reviewed.

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